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COMPLETE SPECIFICATION
(See section 10 and M.13)

SELF-HEALING GROCERY BAG WITH ENHANCED DURABILITY

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**The following specification particularly describes the invention and the manner in
which it is to be performed**

SELF-HEALING GROCERY BAG WITH ENHANCED DURABILITY

Field of the Invention

This invention relates to grocery bags, specifically to reusable bags made from materials with self-healing properties to enhance durability and longevity.

Background of the Invention

Conventional grocery bags, whether disposable or reusable, are prone to tears and punctures during use. This reduces their functional lifespan and contributes to environmental waste. There is a need for a durable, reusable bag that can self-repair minor damage to extend its usability and reduce waste.

US20140291329 - Reusable and disposable grocery bags. Embodiments described herein are related to a reusable grocery bag with a set of disposable liners coupled to an inner surface of the reusable grocery bag. Embodiments disclosed herein utilize disposable liners to reduce, limit or eliminate the spread of bacteria and germs while a consumer is shopping for various items. In embodiments, the disposable liner may be comprised of biodegradable materials.

Summary of the Invention

The present invention provides a reusable grocery bag (100) made from a self-healing material. The bag (100) integrates advanced polymeric or bio-based materials that can autonomously repair small tears, cuts, or punctures when subjected to mechanical or thermal stimuli.

The self-healing process is facilitated by intrinsic polymer bonds or embedded microcapsules (102) containing a healing agent. Upon damage, these mechanisms activate, sealing the tear without requiring external intervention. The bag (100) retains its structural integrity and load-carrying capacity after multiple self-healing cycles.

Detailed Description of the Invention

A durable, flexible polymer matrix, such as polyurethane or bio-derived elastomers, integrated with self-healing properties. Embedded fibres or threads for additional strength and load distribution. Utilizes dynamic covalent bonds (e.g., Diels-Alder reactions) or hydrogen bonds for reformation after damage. Intrinsic systems rely on the material's inherent chemical bonds that can reform after being disrupted. These bonds, such as Diels-Alder reactions, are designed to break under damage and reform upon the application of heat or pressure.

The polymer chains reconnect through reversible weak bonds, ensuring a seamless repair without requiring additional substances. Polymers with ionic groups allow healing by re-establishing ionic bonds.

In the extrinsic mechanism the microcapsules (102) or vascular networks containing healing agents that are released upon rupture. Extrinsic systems incorporate additional elements into the material, such as:

Microcapsules (102) tiny capsules embedded in the material contain a liquid healing agent, such as resin or adhesives. When the bag (100) is damaged, the capsules break open, and the released agent fills and seals the tear. The **vascular networks** are channels built within the material, holding healing agents that are released to the damaged area when the material is ruptured.

The Base Material consists of Polyurethane Matrix with Dynamic Covalent Bonds. The base of the grocery bag (100) is made from polyurethane (PU). Polyurethane is flexible, durable, and exhibits excellent resistance to wear and tear. The polymer is chemically modified with dynamic covalent bonds (e.g., Diels-Alder reactions). These bonds can break under mechanical stress (like a tear) and reform when mild heat is applied, allowing the bag (100) to self-heal. The functionality when a tear occurs, the heat from ambient conditions (e.g., sunlight) or warm water activates the Diels-Alder bonds, causing the polymer chains to reconnect and close the damage.

The extrinsic healing is achieved by Microcapsules (102) with Adhesive Resin. The polyurethane matrix is embedded with microcapsules (102) containing a two-part healing agent: Adhesive Resin (e.g., epoxy or silicone-based resin): This seals and bonds the damaged area. Catalyst or Hardener: Ensures the resin solidifies quickly upon exposure.

The microcapsules (102) are made of biodegradable shells (e.g., urea-formaldehyde or natural polymers) that rupture upon mechanical damage. When the bag (100) is punctured or torn, the microcapsules (102) break open at the damaged site. The adhesive resin flows into the tear and reacts with the hardener, solidifying within minutes to seal the damage. This extrinsic healing ensures quick and efficient repairs even for significant damage.

The polyurethane matrix and microcapsule (102) shells are designed to be recyclable, minimizing environmental impact. Optional use of bio-based fiber (e.g., jute, hemp) and biodegradable polymers ensures that the bag (100) remains eco-friendly at the end of its lifecycle. The intrinsic healing mechanism can be activated at least 50 times without significant degradation in material properties, ensuring long-term reusability.

The combination of intrinsic and extrinsic systems ensures that tears of various sizes can be repaired without user intervention beyond minor activation (e.g., heat or pressure). Reinforcement layers and durable polymers ensure the bag (100) can handle (101) everyday use and heavy loads. Recyclable and biodegradable options reduce the environmental footprint of the bag (100). Quick self-healing activation makes the bag (100) practical for consumers, requiring minimal effort.

Design Features of the bag is with a load Capacity to withstand weights of up to 15 kg without permanent deformation. Ergonomic handles (101) with reinforced edges for comfortable carrying. Compatible with common cleaning agents without compromising the self-healing functionality.

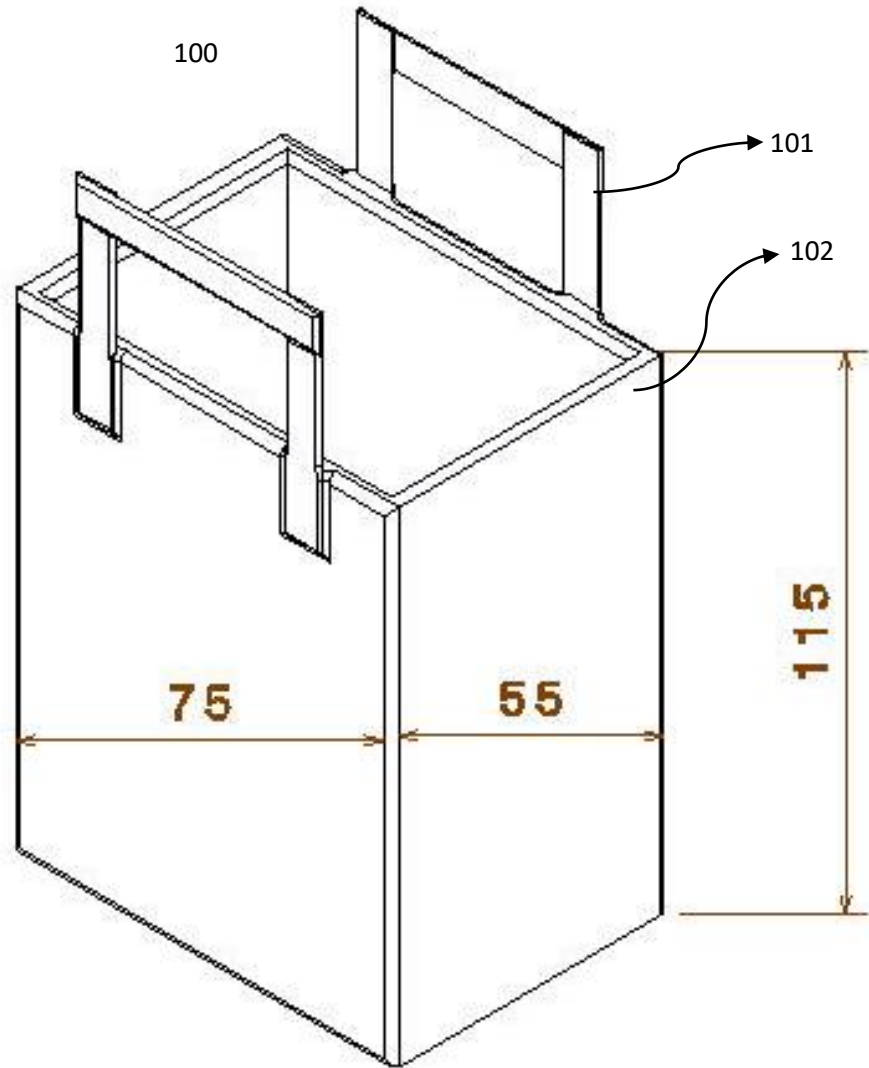
Brief Description of the Drawing:

The rectangular grocery bag (100) with self-healing functionality features two pairs of ergonomic handles (101) on opposite sides, extending vertically from the bag's (100) upper edge, designed for convenient carrying. The main body of the bag (100) appears to have a sturdy structure, with self-healing material, offering ample storage capacity. The design emphasizes simplicity, durability, and utility, suitable for repeated use. The self-healing material is used internally and externally.

Applicant Name: New Horizon College of Engineering

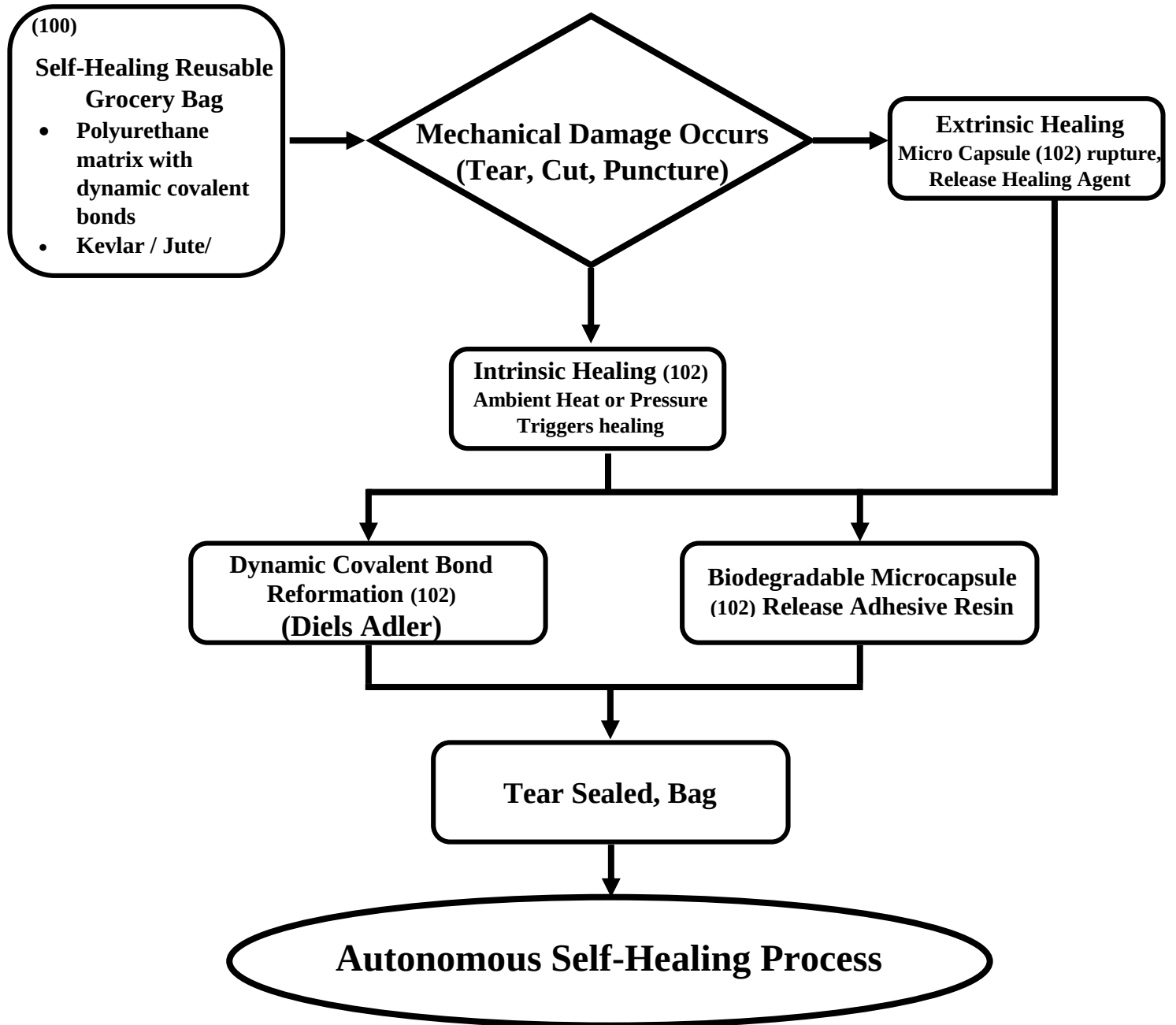
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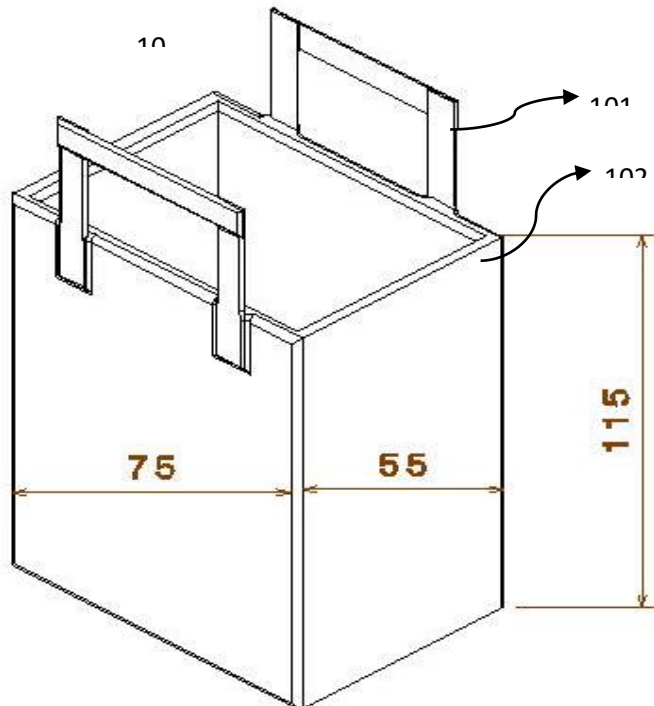
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Title: SELF-HEALING GROCERY BAG WITH ENHANCED DURABILITY

Abstract

The present invention relates to a reusable grocery bag (100) made from an advanced self-healing material that autonomously repairs minor damages such as tears, punctures, or cuts, thereby extending its usability and durability. The bag (100) is constructed using a composite material comprising a polyurethane matrix embedded with self-healing properties. This material employs a dual mechanism: intrinsic healing through dynamic covalent bonds, which reform under thermal or mechanical activation, and extrinsic healing facilitated by microcapsules (102) containing adhesive resin. These microcapsules (102) rupture upon damage, releasing the healing agent to seal larger cuts. Reinforcement fiber, such as Kevlar or biodegradable alternatives, are embedded within the structure to enhance load-bearing capacity and prevent tear propagation. Additionally, a protective silicone-based coating ensures waterproofing, flexibility, and hygiene. This innovation provides a sustainable and durable solution for reusable bags (100), reducing waste and improving consumer convenience by enabling multiple healing cycles with minimal user intervention.



Date: 28/09/2024

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We Claim

1. A grocery bag (100) of claim 1, wherein the polyurethane matrix contains dynamic covalent bonds formed via Diels-Alder reactions, enabling bond reformation under heat application at a temperature range of 50–70°C.
2. The grocery bag (100) of claim 1, wherein the microcapsules (102) are made of a biodegradable polymer shell, such as urea-formaldehyde, and are filled with:
 - a. An epoxy resin or silicone-based adhesive as the healing agent; and
 - b. A corresponding hardener or catalyst for rapid solidification upon exposure to air or pressure.
3. The grocery bag (100) of claim 1, wherein the embedded reinforcement fiber are selected from:
 - a. Kevlar fiber for enhanced strength and resistance to tearing; or
 - b. Biodegradable fiber such as jute or hemp, providing eco-friendly structural reinforcement.
4. The grocery bag (100) of claim 1, further comprising a silicone-based outer coating infused with antimicrobial agents, such as silver nanoparticles or copper oxide, to inhibit bacterial and fungal growth while maintaining flexibility.
5. The grocery bag (100) of claim 1, wherein all materials, including the polyurethane matrix, microcapsule (102) shells, and reinforcement fiber, are recyclable or biodegradable, ensuring an environmentally friendly design.
6. The grocery bag (100) of claim 1, wherein the healing process is activated by: Thermal activation, through exposure to sunlight or warm water for intrinsic healing; and Mechanical activation, by applying pressure to the damaged area for extrinsic healing.
7. The grocery bag (100) of claim 1, wherein the self-healing mechanism is capable of being activated for at least 50 cycles without significant degradation of the material's structural integrity.
8. The grocery bag (100) of claim 1, wherein the composite material construction supports a load of up to 15 kilograms, with reinforcement fiber distributing the weight evenly to prevent strain-induced damage.
9. The grocery bag (100) of claim 1, suitable for: Regular grocery use involving heavy or sharp objects; Long-term reuse due to its enhanced durability and self-repairing capabilities; Sustainable waste reduction through its biodegradable or recyclable material composition.